LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Practice Pack: Matter in Our Surroundings

Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow

For teacher use only. Do not distribute to students.

LEVEL 1: CONCEPTS

Q1. Give two characteristics of the particles of matter.

Answer: (Any two) Particles of matter have spaces between them, are continuously moving, and attract each other.

Q2. Define melting point and boiling point.

Answer: Melting point: the temperature at which a solid melts into a liquid at atmospheric pressure (ice: 273 K). Boiling point: the temperature at which a liquid starts boiling at atmospheric pressure (water: 373 K).

Q3. What is sublimation? Give one example.

Answer: The change of a solid directly into a gas without becoming a liquid (and back). Examples: camphor, naphthalene, ammonium chloride.

Q4. What is latent heat of vaporisation?

Answer: The heat energy needed to change 1 kg of a liquid into gas at atmospheric pressure at its boiling point, without any rise in temperature.

Q5. Convert to the Celsius scale: (a) 300 K (b) 573 K. Convert to the Kelvin scale: (c) 25 °C (d) 373 °C.

Answer: (a) 27 °C (b) 300 °C (c) 298 K (d) 646 K

LEVEL 2: BOARD-STYLE QUESTIONS

Q6. Why does evaporation cause cooling? Give one everyday example.

Answer: During evaporation, the particles of the liquid take energy from the surroundings (or from the surface they are on) to change into vapour, so the surroundings cool down. Examples: we feel cool when sweat evaporates; water in an earthen pot (matka) stays cool.

Q7. Give four factors that affect the rate of evaporation.

Answer: (1) Surface area: more area, faster evaporation. (2) Temperature: higher temperature, faster. (3) Humidity: more humidity, slower. (4) Wind speed: more wind, faster.

Q8. Why do we wear cotton clothes in summer?

Answer: Cotton absorbs sweat well and lets it evaporate. Evaporation takes heat from the body and keeps us cool.

Q9. Why does a gas fill the whole container it is kept in?

Answer: Gas particles have very weak forces of attraction and move very fast in all directions, so they spread out to fill all the available space.

Q10. Why does steam cause more severe burns than boiling water at the same temperature?

Answer: Steam at 373 K carries extra energy (the latent heat of vaporisation) compared with water at 373 K. When it condenses on the skin, it releases this extra heat as well.

LEVEL 3: CHALLENGE (FOUNDATION)

Q11. Why does a desert cooler cool better on a hot, dry day than on a humid day?

Answer: A cooler works by evaporating water. On a dry day the air has little water vapour, so water evaporates quickly and takes away more heat. On a humid day the air already holds a lot of water vapour, so evaporation, and therefore cooling, is slower.

Q12. Water droplets appear on the outside of a glass holding ice-cold water. Where do they come from?

Answer: Water vapour in the air touches the cold surface of the glass, loses energy, and condenses into liquid droplets. The water does not come through the glass.

Q13. Arrange in increasing order of the forces of attraction between particles: water, sugar, oxygen. Explain.

Answer: Oxygen (gas) < water (liquid) < sugar (solid). Particles in solids attract each other most strongly, liquids less, and gases the least.